



9.3. Covariance and Correlation in Data

Previous Section:

 [9.2. Data Distribution, Skewness, and Kurtosis](#)

Contents:

[1. What is Covariance?](#)

[Covariance Matrix](#)

[2. What is Correlation?](#)

[Correlation Matrix](#)

[3. Example: Covariance and Correlation Heatmaps](#)

[Summary](#)

[References](#)

Covariance and correlation are two important statistical measures used in Exploratory Data Analysis (EDA) to examine the relationship between numerical variables. They help determine whether variables tend to increase or decrease together and how strong those relationships are.

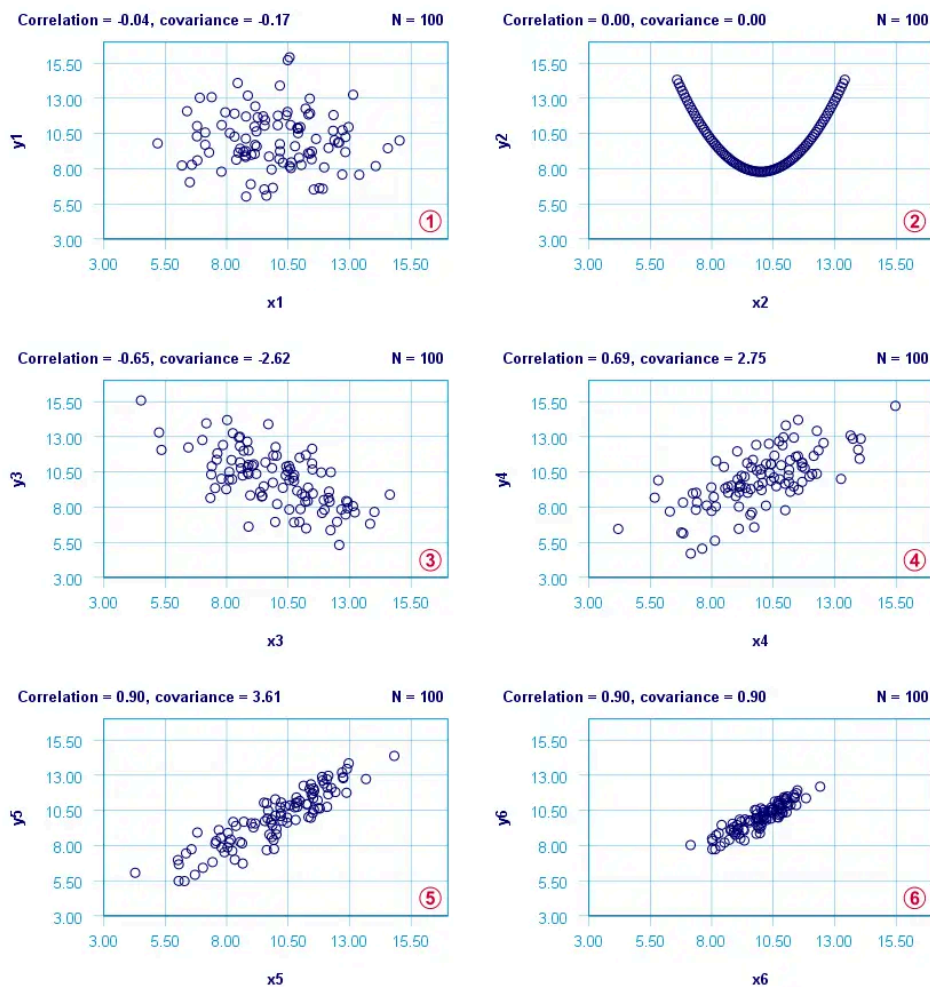
While covariance measures the direction of the relationship, correlation standardizes the measurement, making it easier to compare relationships across different variables.

This section only covers these concepts in the Data Analytics standpoint so there are no formulas or example calculations. If you want a section with those, go and read the Covariance Matrix and Correlation Matrix in the Machine Learning course:

[**Covariance Matrix and Correlation Matrix**](#)

1. What is Covariance?

Covariance measures how two variables vary together. It indicates whether changes in one variable are associated with changes in another.



- **Positive covariance (> 0):** Both variables tend to increase or decrease together.
- **Negative covariance (< 0):** One variable tends to increase while the other decreases.
- **Covariance close to 0:** Little or no linear relationship exists between the variables.

Unlike correlation, the magnitude of covariance depends on the units of the variables, making it difficult to compare relationships across different datasets.

Covariance Matrix

A **Covariance Matrix** is a square matrix that displays the covariance between every pair of numerical features in a dataset.

For example, with three features (Age, Height, Weight), the covariance matrix is:

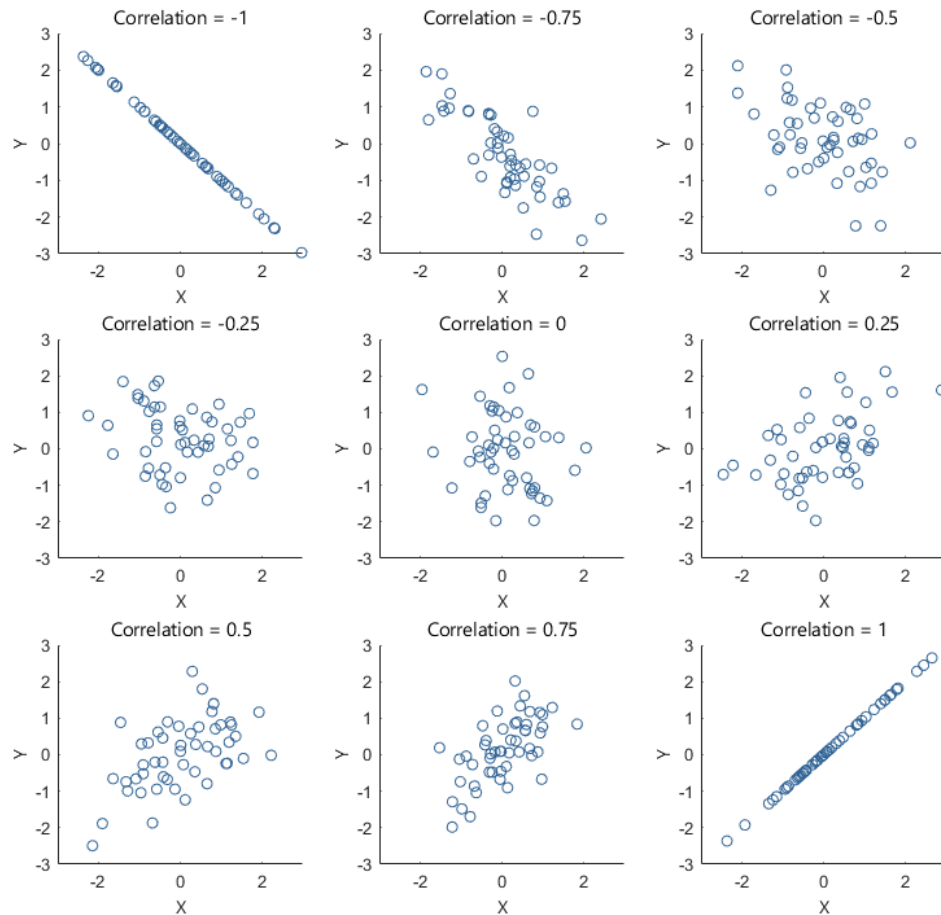
	Age	Height	Weight
Age	Var(Age)	Cov(Age, Height)	Cov(Age, Weight)
Height	Cov(Height, Age)	Var(Height)	Cov(Height, Weight)
Weight	Cov(Weight, Age)	Cov(Weight, Height)	Var(Weight)

The diagonal elements contain the variances of each feature, while the off-diagonal elements represent pairwise covariances.

2. What is Correlation?

Correlation measures both the **strength** and **direction** of the linear relationship between two variables. Unlike covariance, it is standardized, allowing relationships between different pairs of variables to be compared directly.

**Realizations of couples of random variables X and Y
with different correlation coefficients**



The **Pearson correlation coefficient** (r) is defined within the range $-1 \leq r \leq 1$ where:

- $r > 0$: Positive linear relationship (both variables increase together).
- $r < 0$: Negative linear relationship (one increases while the other decreases).
- $r = 0$: No linear relationship.

The closer the coefficient is to ± 1 , the stronger the linear relationship. Values near **0** indicate weak or no linear relationship.

Correlation Matrix

A **Correlation Matrix** is the standardized version of the covariance matrix. Every value represents the correlation coefficient between two features.

For a dataset with **n features**, the correlation matrix has dimensions **n × n**.

- Each row and column represents one feature.
- Every cell shows the correlation between two features.
- The diagonal values are always **1**, since every variable is perfectly correlated with itself.

Important Note: Both the **Covariance Matrix** and the **Correlation Matrix** are **symmetric matrices**, meaning $Cov(X, Y) = Cov(Y, X)$ and $Corr(X, Y) = Corr(Y, X)$. Therefore, only half of the matrix contains unique information.

3. Example: Covariance and Correlation Heatmaps

```
import numpy as np
import pandas as pd

np.random.seed(42)

n = 100

age = np.random.randint(18, 31, n)
height = 145 + age * 1.5 + np.random.normal(0, 4, n)
weight = 20 + height * 0.38 + np.random.normal(0, 5, n)

df = pd.DataFrame({
    "Age": age,
    "Height": np.round(height, 1),
    "Weight": np.round(weight, 1)
})

df.head()
```

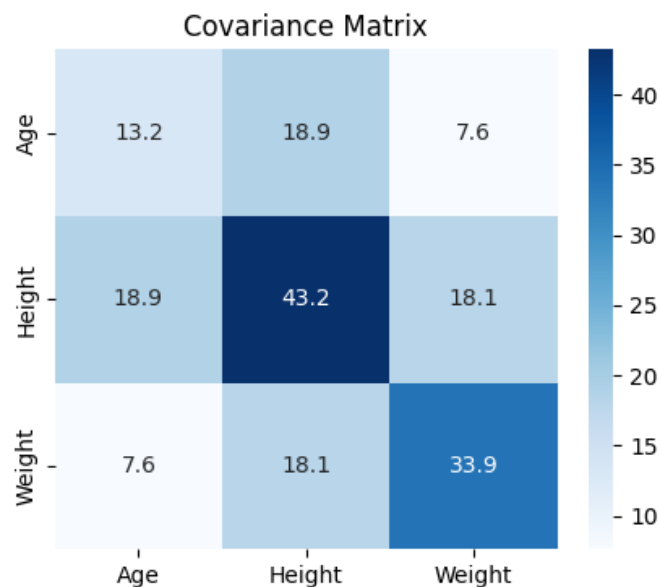
- Covariance Matrix

```
cov_matrix = df.cov()
print(cov_matrix)
```

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(5,4))
sns.heatmap(cov_matrix,
            annot=True,
            cmap="Blues",
            fmt=".1f")

plt.title("Covariance Matrix")
plt.show()
```



- Correlation Matrix

```
corr_matrix = df.corr()
print(corr_matrix)
```

```
plt.figure(figsize=(5,4))
sns.heatmap(corr_matrix,
            annot=True,
            cmap="coolwarm",
            vmin=-1,
            vmax=1,
            fmt=".2f")

plt.title("Correlation Matrix")
plt.show()
```

Summary

Covariance and correlation are both used to study relationships between numerical variables.

- Covariance indicates whether variables move in the same or opposite directions, but its magnitude depends on the measurement units.
- Correlation standardizes this relationship to a scale between **-1 and 1**, making it much easier to interpret and compare across variables.
- In practice, **correlation matrices and heatmaps** are widely used in Exploratory Data Analysis to quickly identify strong positive or negative relationships, detect redundant features, and guide feature selection before building machine learning models.

References

<https://www.geeksforgeeks.org/data-analysis/mathematics-covariance-and-correlation/>

<https://www.datacamp.com/blog/covariance-vs-correlation>

<https://builtin.com/data-science/covariance-vs-correlation>

Next Section:

 [9.4. Outliers in Data](#)